

Monetary Economics and Policy

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Chapter 9: Liquidity Trap



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What We Will Learn

- ▶ How a negative demand shock coupled with the **Zero Lower Bound (ZLB)** creates a liquidity trap and a shortage of aggregate demand.
- ▶ How **Helicopter Money** and unconventional asset-purchase programs (**QE**) can reflate the price level and stimulate consumption through explicit or implicit wealth transfers.
- ▶ Why pure asset swaps are neutral unless accompanied by an appropriate remittance policy that alters private-sector net worth.
- ▶ The role of **Forward Guidance** and price-level targeting in overcoming the ZLB, and how history dependence affects the optimal path of inflation and output.
- ▶ How to manage central bank **reserves** in a liquidity trap, when taxes are distortionary, and how the central bank acts as a **Buyer of Last Resort** to ensure financial stability.

Outline

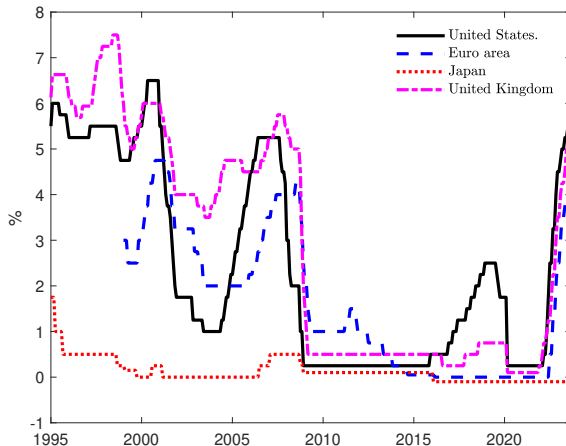
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Introduction

- ▶ **Liquidity traps** describe a situation in which an economy remains in a slump due to a shortage of aggregate demand, even when the nominal interest rate is zero.
- ▶ **Hicks 1937: The “Economics of Depression”**
 - ▶ Describes a scenario where the *LM* curve becomes horizontal.
 - ▶ Any liquidity the authorities inject is absorbed without affecting either the interest rate or income.
- ▶ **The modern perspective**
 - ▶ In modern central banking, the money supply is **endogenous** given the policy rate.
 - ▶ Nevertheless, a zero floor on the interest rate exists because money can circulate in **physical form**.
 - ▶ At the zero lower bound (ZLB), the central bank loses its standard policy instrument and must resort to unconventional measures.

Evolution of Policy Rates



Policy rates for the Euro Area, Japan, U.K., and U.S. (Jan 1995 – Oct 2023). Source: Thomson Reuters Datastream.

The Global Experience with the Zero Lower Bound

- ▶ The **policy rate has hit the ZLB** in several advanced economies.
- ▶ Japan entered a period of stagnation following the collapse of the asset price bubble in 1991.
- ▶ Policy rates have remained below 1% between 1995 and 2024, reaching even negative territory (−10 bps).

The Global Financial Crisis and Beyond

- ▶ United States entered the ZLB in December 2008, maintaining quasi-zero rates for seven years.
- ▶ Euro Area shifted toward the ZLB significantly in 2014 following the sovereign debt crisis.
- ▶ Major central banks began a sharp **liftoff** in 2022 to combat the global surge in inflation, including Japan since 2024.

Analytical Framework: The Euler Equation

- ▶ To see how a demand shock and the ZLB create a shortage of aggregate demand, we consider the *AD* equation in its non-linear Euler-equation version (5.10) under perfect foresight:

Euler Equation

$$U_c(C_t) = \beta(1 + i_t) \frac{P_t}{P_{t+1}} \frac{\xi_{t+1}}{\xi_t} U_c(C_{t+1}) \quad (9.1)$$

Interpretation

- ▶ C_t, i_t, P_t : Consumption, nominal interest rate, and price level.
- ▶ $\beta \in (0, 1), \xi_t$: Subjective discount factor, preference shock.
- ▶ $U_c(\cdot)$: Marginal utility.

Short Run and Long Run Assumptions

Time Horizon

- ▶ Let t_0 denote the **short run** and $t_0 + 1$ the **long run**, characterized by two fundamental assumptions:

1. Nominal Rigidities

- ▶ **Short Run** (t_0): Prices are fully rigid at $P_{t_0} = \bar{P}$.
- ▶ **Long Run** ($t_0 + 1$): Prices are flexible at $P_{t_0+1} = P^*$, determined endogenously by monetary and fiscal policy.

2. Preference Shock

- ▶ The preference shock is low in the short run ξ_{t_0} and high in the long run $\xi_{t_0+1} = \bar{\xi}$, such that $\xi_{t_0} < \bar{\xi}$.
- ▶ Since ξ is a multiplicative shock to the utility flow, a higher $\bar{\xi}$ implies more patient agents, creating an incentive to increase savings.

Goods Market Clearing

Goods Market Clearing

- ▶ In the long run ($t_0 + 1$), the absence of nominal rigidities ensures that the goods market clears.
- ▶ Consumption is determined by the **natural rate of output** (Y^n).
- ▶ For analytical convenience, we assume the long run natural rate of output to be constant, i.e. $Y_{t+1}^n = Y$, implying:

$$C_{t_0+1} = Y$$

Short-Run Aggregate Demand

- ▶ Substituting the information above into the Euler Equation (9.1), we obtain:

$$U_c(C_{t_0}) = \beta(1 + i_{t_0}) \frac{\bar{P}}{P^*} \frac{\bar{\xi}}{\xi_{t_0}} U_c(Y) \quad (9.2)$$

Interpretation

- ▶ Since $U_c(\cdot)$ is a decreasing function of its argument, C_{t_0} depends **negatively** on the nominal rate i_{t_0} .
- ▶ The term $\frac{\bar{\xi}}{\xi_{t_0}} > 1$ represents a **negative demand shock**. It puts upward pressure on $U_c(C_{t_0})$, which forces current consumption C_{t_0} to fall.
- ▶ To maintain $C_{t_0} = Y$ in the face of a patience shock ($\xi_{t_0} \downarrow$), the central bank must lower the nominal interest rate i_{t_0} to offset the pressure for increased saving.

Log-linear Version

Isoelastic Utility Assumption

- Assuming $U(C) = \frac{C^{1-\sigma^{-1}}}{1-\sigma^{-1}}$, with $\sigma > 0$, log-linearization of equation (9.2) yields:

$$c_{t_0} = y - \sigma \left(i_{t_0} - (p^* - \bar{p}) - r_{t_0}^e \right), \quad (9.3)$$

where the frictionless rate $r_{t_0}^e$ is defined by preference shocks:

$$r_{t_0}^e = -\ln \beta + \ln \xi_{t_0} - \ln \bar{\xi}.$$

- A low realization of ξ_{t_0} (increased patience) implies a fall in $r_{t_0}^e$.
- Notation:** Lowercase letters are logs and $i_{t_0} \approx \ln(1 + i_{t_0})$.

The Liquidity Trap Condition

- ▶ If the negative shock to preferences is large enough to drive the frictionless real rate to a large negative value such that:

$$r_{t_0}^e + (p^* - \bar{p}) < 0, \quad (9.4)$$

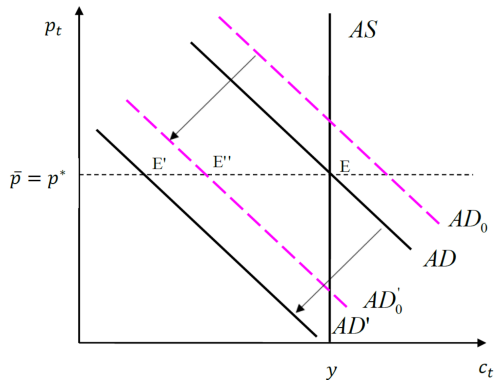
the economy suffers a persistent shortage of aggregate demand, $(c_{t_0} < y)$, even at the zero floor $(i_{t_0} = 0)$, as implied by (9.3).

- ▶ **Result:** The economy finds itself in a **liquidity trap**.

Graphical Analysis: Preference Shock

The Initial State E

- ▶ **Supply Side:** The AS curve is vertical at constant output y .
- ▶ **Demand Side:** The initial AD crosses AS at price $\bar{p}$ and consumption y .
- ▶ **Policy:** The nominal rate is positive ($i_{t_0} > 0$) and the long-run price anchor is met ($\bar{p} = p^*$).
- ▶ **The ZLB Schedule:** AD_0 displays AD when $i_{t_0} = 0$.



Graphical Analysis: Dynamics of the Slump

Preference Shock and Initial Contraction ($E \rightarrow E'$)

- ▶ A preference shock ($\xi_{t_0} \downarrow$) increases the marginal utility of future consumption, raising the desire to save.
- ▶ This shifts the aggregate demand schedules left.
- ▶ Due to short-run price rigidity at $\bar{p}$, the economy moves to equilibrium E' , where consumption falls strictly below potential ($c_{t_0} < y$).

Monetary Policy Response and the Trap ($E' \rightarrow E''$)

- ▶ The central bank can attempt to offset the demand deficiency by lowering i_{t_0} , shifting AD' at most to AD'_0 .
- ▶ **The Liquidity Trap:** The new equilibrium is E'' . Despite $i_{t_0} = 0$, the real interest rate remains too high to clear the goods market, leaving the economy in a persistent slump.

Prerequisites for a Liquidity Trap

Three Necessary Conditions

- ▶ **Negative Demand Shock:** Preference shifts ($\xi \downarrow$) that push r^e into negative territory.
- ▶ **Price Rigidities:** Prevents the price level from falling to clear the goods market.
- ▶ **Zero Lower Bound (ZLB):** A floor on nominal rates due to the existence of physical currency.

No Liquidity Trap occurs under:

- ▶ **Flexible Prices:** The goods market clears even in the short run via price adjustment.
- ▶ **No ZLB Constraint:** The central bank sets $i_{t_0} < 0$ to bridge the demand gap.

Note: The Real Rate and Inflation Expectations

The Real Rate Disconnect

- ▶ Real rate definition: $r_{t_0} = i_{t_0} - (p^* - \bar{p})$.
- ▶ A liquidity trap arises when the actual real rate is “too high” relative to the frictionless rate: $r_{t_0} > r_{t_0}^e$.
- ▶ This disconnect creates **excess saving**, depressing current consumption and aggregate demand.

Exiting the Trap: The Role of Inflation Expectations

- ▶ Since i_{t_0} is constrained by the ZLB, the real rate can also fall to align with the frictionless rate through a rise in **inflation expectations**, i.e. p^* goes up.

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Outline of the Results

This chapter studies tools to escape the liquidity trap when conventional monetary policy is constrained. The main results are:

- ▶ **Helicopter Money:** Direct transfers to the private sector stimulate aggregate demand through a **wealth effect**, effectively increasing consumption and inflation.
- ▶ **Asset Purchase Programs:** By absorbing private sector risk, these programs generate **implicit transfers** that bolster net worth and elevate inflation expectations.
- ▶ **Forward Guidance:** Managing future policy rate trajectories to boost inflation expectations and reduce current/future real rates.
- ▶ **Reserve Management:** The optimal accumulation and withdrawal of central bank reserves to manage liquidity when taxation is distortionary.
- ▶ **Lender of Last Resort:** Mitigating systemic risk by providing liquidity against illiquid assets to alleviate credit market constraints.

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Helicopter Money: Theoretical Framework

The Friedman (1969) Paradigm

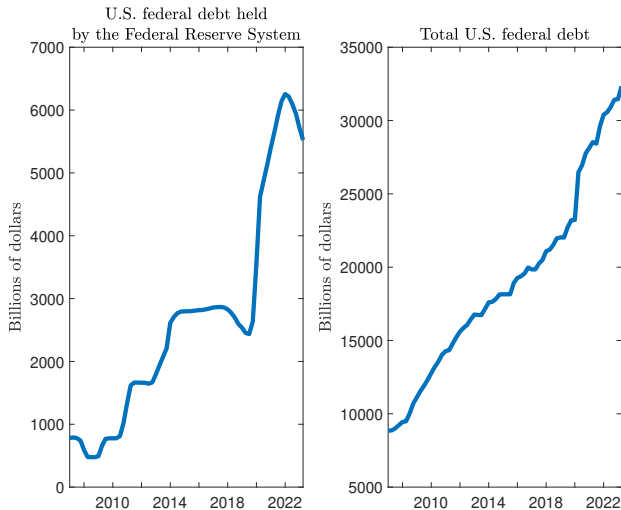
- ▶ A unique, non-recurrent event where currency is dropped to the community.
- ▶ **Monetary Logic:** Deflation is a state where the price of currency is too high. Since central banks print default-free liabilities, they can **reflate** the economy by increasing the supply of these units as a permanent "gift".

Transmission Mechanism

- ▶ **Reflation:** Stimulates demand by raising long-run price expectations (p^*), which lowers current real interest rates.
- ▶ **Operational Step:** The Treasury issues debt to fund private-sector transfers, which the central bank monetizes via permanent reserve expansion.
- ▶ **The Wealth Effect:** Unlike standard debt-financed transfers, this injection is not offset by future tax liabilities, leading to a permanent increase in private nominal net worth.

U.S. Federal Debt and Federal Reserve Holdings

- **Coordination:** Episodes of fiscal-monetary alignment have been frequent since 2007. The surges in federal debt and Federal Reserve holdings have mirrored each other.
- **Monetization:** This could represent the operational outcome of a “helicopter drop”.



Source: St. Louis Fed.

United States: Crisis Interventions

2007–2008 Financial Crisis

- ▶ **Fiscal:** American Recovery and Reinvestment Act (\$787bn stimulus) launched in February 2009.
- ▶ **Monetary:** QE1 launched (\$300bn in long-term debt and MBS) in March 2009; QE2 expanded purchases by \$600bn in November 2010.

COVID-19 Pandemic Relief (March 2020 – March 2021)

- ▶ **Fiscal:** Five stimulus packages totaling over \$5tn, featuring direct cash transfers, tax credits, and expanded unemployment benefits.
- ▶ **Monetary:** The Federal Reserve accumulated $\approx$ \$3tn in Treasury securities and implemented direct/indirect lending to small and medium-sized businesses.

Europe: Pandemic Emergency Purchase Programme (PEPP)

ECB Monetary Expansion & Fiscal Coordination

- ▶ Activation of PEPP at €1.8 trillion, purchasing sovereign debt of euro-area national treasuries.
- ▶ *Note:* Unlike the 2015 QE, this was explicitly linked to a coordinated fiscal expansion.
- ▶ The European Commission relaxed the 3.0% deficit rule, allowing for substantial national deficits.
- ▶ Implementation of a €800 billion European-wide fiscal stimulus.

Model

Consider a representative household's intertemporal utility including also real money balances (see Chapter 2):

$$\sum_{t=t_0}^{\infty} \beta^{t-t_0} \xi_t \left[U(C_t) + V\left(\frac{M_t}{P_t}\right) \right].$$

Interpretation

- ▶ M_t : Cash.
- ▶ $V(\cdot)$: utility in real money balances M_t/P_t .
- ▶ $V(\cdot)$ has a satiation point at $\bar{m} > 0$:

$$V_m\left(\frac{M_t}{P_t}\right) = 0 \quad \text{for} \quad \frac{M_t}{P_t} \geq \bar{m}.$$

Structural Assumptions

- ▶ **Preference Shock:** $\xi_t = \bar{\xi}$ for each $t \geq t_0 + 1$ while $\xi_{t_0} \leq \bar{\xi}$.
- ▶ **Short Run (t_0):** Prices are fixed at $\bar{P}$.
- ▶ **Long Run ($t \geq t_0 + 1$):** Prices are fully flexible.
- ▶ **Equilibrium:** Price flexibility and constant endowment imply $C_t = Y$ for each $t \geq t_0 + 1$.

Short-Run Equilibrium Conditions

- Using the above assumptions into the Euler equation (9.1):

Euler Equation

$$U_c(C_{t_0}) = \beta(1 + i_{t_0}) \frac{\bar{P}}{P_{t_0+1}} \frac{\bar{\xi}}{\xi_{t_0}} U_c(Y). \quad (9.5)$$

- As discussed in Chapter 2, equilibrium in the money market implies at time t_0 :

$$\frac{M_{t_0}}{\bar{P}} \geq L(C_{t_0}, i_{t_0}).$$

- **At the ZLB:** When $i_{t_0} = 0$, agents are indifferent between holding money and bonds. The money supply M_{t_0} is absorbed with the inequality holding strictly.

Long-Run Price Determination

- ▶ The price level P_{t_0+1} is endogenously determined by the equilibrium conditions discussed in **Chapter 2**. We restate them:

The Fisher Equation

- ▶ Relates the nominal rate to the real rate ($1/\beta$) and inflation:

$$1 + i_t = \frac{1}{\beta} \frac{P_{t+1}}{P_t}. \quad (9.6)$$

Money Market Equilibrium

- ▶ Real balances satisfy the liquidity preference relationship:

$$\frac{M_t}{P_t} \geq L(Y, i_t), \quad (9.7)$$

holding with equality whenever $i_t > 0$, Note that consumption is equal to Y in the long run.

Intertemporal Budget Constraint

- ▶ Following the household's optimization in **Chapter 2**, the intertemporal budget constraint must be satisfied in equilibrium:

Intertemporal Budget Constraint

$$\sum_{t=t_0+1}^{\infty} \beta^{t-t_0-1} \left(\frac{T_t}{P_t} + \frac{i_t}{1+i_t} \frac{M_t}{P_t} \right) = \frac{B_{t_0} + X_{t_0} + M_{t_0}}{P_{t_0+1}}. \quad (9.8)$$

Interpretation

- ▶ B_{t_0} : Treasury bonds at time t_0 .
- ▶ X_{t_0} : central bank reserves at time t_0 .

Central Bank Budget Constraint

Flow Budget Constraint

$$\frac{B_t^C - X_t}{1 + i_t} - M_t = B_{t-1}^C - X_{t-1} - M_{t-1} - T_t^C. \quad (9.9)$$

Interpretation

- ▶ B_t^C : Holdings of short-term, default-free assets (private or treasury).
- ▶ T_t^C : Remittances transferred to the treasury (if $T^C > 0$).

Treasury Budget Constraint

Flow Treasury Budget Constraint

$$\frac{B_t^F}{1+i_t} = B_{t-1}^F - T_t - T_t^C, \text{ where } B_t^F \text{ is treasury debt.} \quad (9.10)$$

Solvency Constraint

- The trajectory of taxes T is constrained by intertemporal solvency:

$$\frac{B_{t_0}^F}{P_{t_0+1}} = \sum_{t=t_0+1}^{\infty} \beta^{t-t_0-1} \left(\frac{T_t}{P_t} + \frac{T_t^C}{P_t} \right). \quad (9.11)$$

Debt Market Clearing

$$B_t^F = B_t + B_t^C. \quad (9.12)$$

Equilibrium Definition

Long-run equilibrium: A set of sequences

$$\{P_t, i_t, X_t, M_t, B_t, B_t^C, B_t^F, T_t, T_t^C\}_{t=t_0+1}^{\infty},$$

with $\{P_t, i_t, X_t, M_t, B_t^C\}_{t=t_0+1}^{\infty} \geq 0$, satisfying, for all $t \geq t_0 + 1$:

- ▶ Fisher equation (9.6)
- ▶ Money market equilibrium (9.7)
- ▶ central bank budget constraint (9.9)
- ▶ Treasury budget constraint (9.10)
- ▶ Bond market clearing (9.12)

and satisfying the private sector intertemporal budget constraint (9.8) given inherited values from period t_0 :

$$B_{t_0}^C, B_{t_0}^F, X_{t_0}, M_{t_0}.$$

The sequence of taxes $\{T_t\}_{t=t_0+1}^{\infty}$ is restricted to satisfy the Treasury solvency condition (9.11), given equilibrium sequences $\{P_t, T_t^C\}_{t=t_0+1}^{\infty}$ and the inherited value $B_{t_0}^F$.

Case 1: Helicopter Money under Coordinated Policy

Monetary and Fiscal Policy Coordination

- ▶ This regime relies on the **backing of Treasury liabilities by the central bank** .
- ▶ The central bank's guarantee on Treasury debt relaxes the standard Treasury solvency constraint (9.11).

Determination of the Price Sequence

Interest-rate Policy Assumption

- For all $t \geq t_0 + 1$, the central bank commits to a constant inflation target $\Pi > \beta$:

$$1 + i_t = \frac{\Pi}{\beta} \implies \frac{P_{t+1}}{P_t} = \Pi,$$

pinning down the path of inflation and interest rates through the Fisher equation (9.6), given P_{t_0+1} .

- Given the interest-rate policy, real money balances (9.7) for $t \geq t_0 + 1$ are constant:

$$\frac{M_t}{P_t} = L\left(Y, \frac{\Pi}{\beta} - 1\right),$$

pinning down the path of money holdings M_t for $t \geq t_0 + 1$, given P_{t_0+1} .

Nominal Anchor

- The level of prices P_{t_0+1} still needs to be determined.

Real Tax Policy Assumption

$$\frac{T_t}{P_t} = (1 - \beta)\tau, \quad \tau > 0.$$

- Substituting the tax policy and seigniorage into the IBC (9.8):

$$\frac{B_{t_0} + X_{t_0} + M_{t_0}}{P_{t_0+1}} = \tau + \mathcal{S}(\Pi, Y),$$

where

$$\mathcal{S}(\Pi, Y) \equiv \sum_{t=t_0+1}^{\infty} \beta^{t-t_0-1} \frac{i_t}{1+i_t} \frac{M_t}{P_t} = \frac{\Pi - \beta}{\Pi(1 - \beta)} L\left(Y, \frac{\Pi}{\beta} - 1\right).$$

Price-Level Determination under Coordination

The Nominal Anchor

$$P_{t_0+1} = \frac{B_{t_0} + X_{t_0} + M_{t_0}}{\tau + \mathcal{S}(\Pi, Y)} \quad (9.13)$$

- ▶ The long run price level is proportional to the outstanding government liabilities carried from the short run (time t_0).
- ▶ Increasing them raises the price level proportionally.

Reflating and Stimulating Aggregate Demand

Policy Objective to Exit the Liquidity Trap

- ▶ The government should **reflate the economy** by raising P_{t_0+1} and **stimulating consumption** C_{t_0} via the Euler equation (9.5).

The "Helicopter Drop" Mechanism

- ▶ A cut in taxes T_{t_0} at time t_0 is financed by increasing nominal liabilities (B_{t_0} , X_{t_0} , or M_{t_0}).
- ▶ **Liability Equivalence:** Since the short-run nominal interest rate is zero ($i_{t_0} = 0$), the composition of these liabilities does not matter.

The Impact on the Price Level

- Consolidating the central bank (9.9) and Treasury (9.10) budgets at t_0 (with $i_{t_0} = 0$):

$$B_{t_0} + X_{t_0} + M_{t_0} = B_{t_0-1} + X_{t_0-1} + M_{t_0-1} - T_{t_0} \quad (9.14)$$

- A lower T_{t_0} (tax cut) increases total nominal liabilities in (9.14).
- From (9.13), if the real backing (τ and $\mathcal{S}$) remains constant, the increase in the numerator leads to a proportional increase in P_{t_0+1} .

Liability Equivalence

Irrelevance of Asset Composition

- ▶ The t_0 tax cut defined in (9.14) can be financed through two equivalent channels:
 - ▶ **Private Sector:** The Treasury issues debt B_{t_0} directly to the public.
 - ▶ **Central Bank:** The Bank issues liabilities X_{t_0} (reserves) to purchase Treasury debt.
- ▶ At the ZLB ($i_{t_0} = 0$), these financing options are perfect substitutes.
- ▶ Large-scale asset purchases produce equivalent effects regardless of whether the program targets short-term or long-term debt instruments.
- ▶ It is analytically irrelevant whether the central bank holds the Treasury debt permanently or writes it off.
- ▶ The economic impact depends solely on the **unbacked** transfer made to the private sector.

Requirements and Mechanism

Permanent Unbacked Liability Expansion

- ▶ Effectiveness depends on a constant denominator $(\tau + S)$ in (9.13).
- ▶ Therefore, the Treasury must commit to **never undo the** tax relief.

Transmission Channel

- ▶ **Wealth Effect:** Increased nominal liabilities (helicopter drop) raise household net wealth.
- ▶ **Excess Demand:** With output Y fixed, stimulated consumption cannot be met by higher production.
- ▶ **Price Adjustment:** The resulting excess demand is fully absorbed by a proportional increase in the price level P_{t_0+1} .

Case 2: Helicopter Money by Central Bank Only

Institutional Setting

- ▶ We consider a regime where the central bank does not back Treasury liabilities, as discussed in Chapter 1, and study whether similar Helicopter money experiments can be undertaken.
- ▶ The Treasury must remain solvent independently, satisfying (9.11).
- ▶ Combining the solvency constraint (9.11) with the intertemporal budget constraint (9.8) yields:

$$\frac{X_{t_0} + M_{t_0} - B_{t_0}^C}{P_{t_0+1}} = \sum_{t=t_0+1}^{\infty} \beta^{t-t_0-1} \left(\frac{i_t}{1+i_t} \frac{M_t}{P_t} - \frac{T_t^C}{P_t} \right). \quad (9.15)$$

The Nominal Anchor

Policy Assumptions

- ▶ **Constant inflation target:** $\Pi > \beta$ and $1 + i_t = \frac{\Pi}{\beta}$.
- ▶ **Constant Remittances:** Nominal policy $\{T_t^C = T^C\}_{t=t_0+1}^{+\infty}$.
- ▶ Inserting these assumptions into (9.15) and using previous results:

$$\frac{X_{t_0} + M_{t_0} + \mathcal{T}^C - B_{t_0}^C}{P_{t_0+1}} = \mathcal{S}(\Pi, Y),$$

where $\mathcal{T}^C$ represents:

$$\mathcal{T}^C \equiv \frac{\Pi}{\Pi - \beta} T^C. \quad (9.16)$$

Price Level Determination

The Long-Run Price Level

- Rearranging the previous expression, we obtain the long-run price level P_{t_0+1} :

$$P_{t_0+1} = \frac{X_{t_0} + M_{t_0} + \mathcal{T}^C - B_{t_0}^C}{\mathcal{S}(\Pi, Y)}.$$

- Alternatively, expressed in terms of central bank net worth (N^C):

$$P_{t_0+1} = \frac{\mathcal{T}^C - N_{t_0}^C}{\mathcal{S}(\Pi, Y)}, \quad (9.17)$$

where $N_{t_0}^C \equiv B_{t_0}^C - X_{t_0} - M_{t_0}$.

- Since $\mathcal{S}(\Pi, Y) > 0$, a positive price level requires:

$$\mathcal{T}^C > N_{t_0}^C.$$

The Mechanism of Reflation

The Mechanism of Reflation

- ▶ To increase P_{t_0+1} , the central bank must reduce its nominal net worth $N_{t_0}^C$ in the numerator of

$$P_{t_0+1} = \frac{\mathcal{T}^C - N_{t_0}^C}{S(\Pi, Y)}, \quad (9.18)$$

ceteris paribus.

- ▶ At the ZLB ($i_{t_0} = 0$), central bank profits are zero, and the law of motion is:

$$N_{t_0}^C = N_{t_0-1}^C - T_{t_0}^C. \quad (9.19)$$

Implementation Channels

The reduction in nominal net worth $N_{t_0}^C$ can be achieved through two equivalent policy actions:

1. Short-run Transfers to Treasury

- ▶ Increasing $T_{t_0}^C$ expands the monetary base to finance lower current taxes.
- ▶ This corresponds to the standard "helicopter money" narrative.

2. Asset Write-offs

- ▶ The Bank writes off private or public securities held from $t_0 - 1$.
- ▶ This triggers a **direct wealth effect** on private balance sheets, bypasses the Treasury, and raises aggregate demand and P_{t_0+1} .

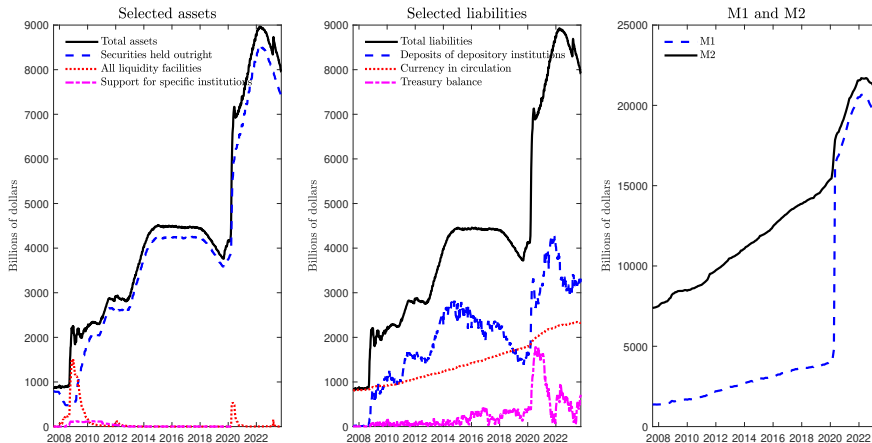
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Unconventional Monetary Policy (UMP)

Institutional Evolution at the Zero Lower Bound

- ▶ **Strategic Shift:** Following the 2008 Financial Crisis and the COVID-19 pandemic, central banks intervened in capital markets through the acquisition of private and government securities with **long-term maturity** and **credit-risk**.
- ▶ **Why "Unconventional"?:** central banks transitioned from holding short-term Treasury debt to actively managing portfolios of mortgage-backed securities, corporate bonds, and long-dated sovereign debt. They also expanded their balance sheets substantially.

Evolution of the FEDs Balance Sheet and Monetary Aggregates



Selected assets of the Federal Reserve System (left panel), selected liabilities (center panel), M1 and M2 (right panel). Billions of dollars. Period: July 2007 - October 2023. Source: Fed Economic Data, St. Louis Fed.

Key Observations

- ▶ **Assets:** Surge in long-term securities and liquidity facilities (e.g., Maiden Lane, Commercial Paper Funding).
- ▶ **Liabilities:** Massive growth in deposits held by depository institutions.
- ▶ **Monetary Aggregates:** Significant surge in M1 and M2, particularly during the pandemic.

Transmission Mechanisms: Two Views

Bernanke (2022) mentions two contrasting views within the FOMC regarding the transmission mechanism of UMP.

1. The Reserve Channel

- ▶ Expanding reserves induces banks to lend more, increasing liquidity and stimulating aggregate demand.

2. The Portfolio Rebalancing Channel

- ▶ Central bank's purchases of long-term securities lower their yields. Investors reallocate portfolios toward other assets, lowering yields across the market.

This Section examines the portfolio-rebalancing channel via a frictionless financial-market model in which securities have only pecuniary benefits.

Environment

Stochastic Environment

- ▶ We extend the model discussed so far to include random preference shocks ξ and output Y .

Asset Characteristics

- ▶ The economy features both short-term (B_S) and long-term (B_L) debt.
- ▶ Market Clearing ensures:

$$\underbrace{B_{S,t}^F = B_{S,t} + B_{S,t}^C}_{\text{Short-Term Bonds}} \quad \text{and} \quad \underbrace{B_{L,t}^F = B_{L,t} + B_{L,t}^C}_{\text{Long-Term Bonds}}$$

where the subscripts F and C denote Treasury issuance and central bank holdings, respectively.

Long-term Bonds

Long-Term Bond Structure

- ▶ Long-term bonds with price Q_t^L at time t pay geometrically decaying coupons $(1, \delta, \delta^2 \dots)$ in subsequent periods.
- ▶ The realized return between $t - 1$ and t on long-term bonds i_t^L is given by:

$$1 + i_t^L = \frac{1 + \delta Q_t^L}{Q_{t-1}^L}$$

Key Implication

- ▶ Due to stochastic shocks, the long-term return may differ from the risk-free short rate ($i_t^L \neq i_{t-1}$).
- ▶ **Note:** The equilibrium is characterized as discussed so far, but now includes an additional asset-pricing condition for Q_t^L .

Irrelevance of Open Market Operations: Experiment

Experiment

- ▶ We assume that the central bank **does not back** the Treasury.
- ▶ Consider a specific sequence of central bank bond holdings:

$$\left\{ \tilde{B}_{L,t}^C, \tilde{B}_{S,t}^C \right\}_{t=t_0}^{\infty}$$

- ▶ **Intervention:** The central bank alters this portfolio (e.g., purchasing more long-term bonds $\tilde{B}_{L,t}^C$) while keeping all other policy specifications unchanged.
- ▶ Does this change the equilibrium allocation and price level?

The Remittance Policy Condition

Policy Specification

- ▶ Constant interest rate at $i_t = 1/\beta - 1$.
- ▶ Exogenous sequence of short and long-term bonds $\{\tilde{B}_{L,t}^C, \tilde{B}_{S,t}^C\}_{t=t_0}^{\infty}$.
- ▶ We assume a specific remittance rule where the central bank transfers a fixed real amount τ^C plus all variable returns:

$$\frac{T_t^C}{P_t} = (1 - \beta)\tau^C + \frac{i_{t-1}M_{t-1} + (i_t^L - i_{t-1})Q_{t-1}^L \tilde{B}_{L,t-1}^C}{P_t}. \quad (9.20)$$

- ▶ **Interpretation:** The central bank rebates seigniorage from money issuance and any excess returns (or losses) from holding long-term assets directly to the Treasury.

Intertemporal Resource Constraint

Equilibrium Intertemporal Resource Constraint

- The relevant condition for determining prices is the stochastic version of the intertemporal resource constraint (9.15), accounting for long-term debt holdings:

$$\frac{X_{t-1} + M_{t-1} - \tilde{B}_{S,t-1}^C - (1 + i_t^L) Q_{t-1}^L \tilde{B}_{L,t-1}^C}{P_t} = E_t \sum_{T=t}^{\infty} R_{t,T} \left(\frac{i_T}{1 + i_T} \frac{M_T}{P_T} - \frac{T_T^C}{P_T} \right), \quad (9.21)$$

where future real flows are discounted by the real stochastic discount factor $R_{t,T}$:

$$R_{t,T} = \beta^{T-t} \frac{\xi_T U_c(Y_T)}{\xi_t U_c(Y_t)}.$$

Simplifying the Equilibrium

Deriving the Price Level Condition

- ▶ To simplify, we assume $\xi_T U_c(Y_T)$ follows a martingale, i.e. $E_t \xi_T U_c(Y_T) = \xi_t U_c(Y_T)$ for each $T > t$. Substituting the transfer rule (9.20) and the martingale property into (9.21), we obtain:

$$\frac{N_{t-1}^C (1 + i_{t-1})}{P_t} = \tau^C.$$

- ▶ Given the policy rate $1 + i_t = 1/\beta$, this simplifies to:

$$\frac{N_{t-1}^C}{P_t} = \beta \tau^C, \tag{9.22}$$

where the net worth N_t^C is defined as:

$$N_t^C = Q_t^L \tilde{B}_{L,t}^C + \frac{\tilde{B}_{S,t}^C - X_t}{1 + i_t} - M_t. \tag{9.23}$$

Equilibrium Price Level Stability

- ▶ Equation (9.22) implies that P_t depends on N_{t-1}^C (predetermined) and parameters, making it **not contingent** on the current state. Using the stochastic Euler equation:

Euler Equation (stochastic version of 9.5)

$$\xi_t U_c(Y_t) = \beta(1 + i_t) E_t \left\{ \frac{P_t}{P_{t+1}} \xi_{t+1} U_c(Y_{t+1}) \right\}.$$

- ▶ Using the martingale property of $\xi_t U_c(Y_t)$, the result that P_{t+1} is not state contingent, and the constant interest rate policy ($1 + i_t = 1/\beta$), it follows that the price level is constant over time:

$$P_t = P^* \quad \text{for all } t \geq t_0.$$

- ▶ It can also be shown that **net worth** remains constant at N_{t_0-1} .

Proof in Appendix

The Irrelevance Result

Equilibrium Characterization

- ▶ With prices determined at $P_t = P^*$, the path of money supply is pinned down by the demand function (9.7) for all $t \geq t_0$.
- ▶ The equilibrium is thus fully defined for the initial policy specification $\{\tilde{B}_{S,t}^C, \tilde{B}_{L,t}^C\}$.

Policy Invariance

- ▶ Consider an alternative balance-sheet policy $\{\bar{B}_{S,t}^C, \bar{B}_{L,t}^C\}$ with active long-term bond purchases ($\bar{B}_{L,t}^C > 0$).
- ▶ Since net worth is anchored at $N_{t_0-1}^C$, the price level P^* and interest rate i_t remain **unchanged**.

Conclusion

- ▶ The equilibrium allocation is independent of whether the central bank holds long-term securities.

The Intuition of Irrelevance

The Necessary Condition for Effectiveness

- ▶ For asset purchases to affect equilibrium prices, they must alter the **total wealth** (financial + human) of households.
- ▶ This is the same mechanism behind "Helicopter Money": creating a wealth effect to stimulate consumption.

Why instead this Policy is Neutral

- ▶ The assumed remittance rule (9.20) creates a closed loop that neutralizes risk:
 - ① **Risk Transfer:** If the central bank incurs losses on risky assets that were before in the hand of households, it reduces transfers to the Treasury.
 - ② **Fiscal Reaction:** To satisfy its solvency constraint, the Treasury must raise lump-sum taxes.
 - ③ **The Offset:** The private sector's reduced financial risk is perfectly offset by increased future tax liabilities. Total private **wealth** does not change.

The Mechanical Accommodation

If prices don't move, what does?

- Constant nominal net worth ($N_t^C = N_{t_0-1}^C$) imposes the following identity:

$$Q_t^L B_{L,t}^C + \frac{B_{S,t}^C}{1 + i_t} - \frac{X_t}{1 + i_t} - M_t = N_{t_0-1}^C$$

- **Endogenous Adjustment:** Any change in the asset mix ($B_{S,t}^C$ vs. $B_{L,t}^C$) is perfectly accommodated by a variation in reserves (X_t), without any change in equilibrium prices Q_t^L , interest rate i_t or money supply M_t .

Conclusion: Limits of Pure Asset Swaps

Summary of Irrelevance

- ▶ Massive asset purchases are ineffective if they function purely as asset swaps without an appropriate transfer policy to generate wealth effects.
- ▶ In the next section, we will see how changing the transfer policy restores effectiveness.

Note: Forward Guidance

- ▶ Our experiment held future interest rate policy constant.
- ▶ However, asset purchases may alter private sector expectations about *future* rates (see the Forward Guidance section), and be indirectly effective through those changes.

Relevance of Unconventional Open Market Operations

- ▶ To break irrelevance, gains or losses must remain on the central bank's balance sheet.
- ▶ We assume a constant **real** remittance policy:

$$\frac{T_t^C}{P_t} = (1 - \beta)\tau^C, \quad \text{with } \tau^C > 0.$$

- ▶ Under this rule (and assuming constant Y, ξ), the budget constraint (9.21) implies:

$$L(Y, \beta^{-1} - 1) + \frac{N_{t-1}^C + \Psi_t^C}{P_t} = \tau^C. \quad (9.24)$$

Portfolio Composition and Price Determination

Impact of Unconventional Interventions

- ▶ Consider an initial equilibrium where the central bank holds only short-term, risk-free assets ($B_{L,t}^C = 0$).
- ▶ Under this specification, profits Ψ_t^C are independent of long-term bond returns.
- ▶ The central bank expands its balance sheet or substitutes assets to hold long-term securities ($B_{L,t}^C > 0$).
- ▶ A negative shock to bond returns reduces profits. To satisfy the budget constraint (9.24) given fixed real remittances τ^C , the price level must adjust downward.
- ▶ **Conclusion:** Under the specified remittances policy, the asset composition of the central bank becomes a determinant of the equilibrium price level.

Sustainability of the Interest Rate Peg

Breakdown of the Peg

- ▶ What happens if the central bank incurs *substantial* losses on its risky portfolio?
- ▶ If these losses are large enough to make the left-hand side of the solvency condition (9.24) negative, the equilibrium breaks down.
- ▶ Under the constant remittance policy τ^C , an interest rate peg of $i = 1/\beta - 1$ (implying zero net inflation) becomes **infeasible**.
- ▶ **Implication:** Without a fiscal recapitalization, the central bank must change policy—abandoning the simple peg and altering its future interest rate trajectory.

Reflation via Seigniorage

Restoring Equilibrium

- ▶ The central bank could target a new nominal interest rate $i = \Pi/\beta - 1$, consistent with a positive inflation target $\Pi > 1$.
- ▶ The intertemporal budget constraint is modified to include **seigniorage revenues** as the balancing mechanism:

$$\underbrace{\frac{\Pi - \beta}{\Pi(1 - \beta)} L\left(Y, \frac{\Pi}{\beta} - 1\right)}_{\text{Seigniorage Revenue}} + \frac{N_{t-1}^C + \Psi_t^C}{P_t} = \tau^C.$$

- ▶ **Mechanism:** A large negative Ψ_t^C (loss) is absorbed by an increase in inflation Π . This generates the seigniorage necessary to cover the shortfall, effectively "reflating" the economy.

Real-World Remittance Policies (e.g., The Fed)

Beyond Constant Remittances

- ▶ In practice, policies are more complex. For instance, the US Federal Reserve rebates profits during normal times but **halts remittances** during periods of loss.
- ▶ Losses are accumulated as a "deferred asset" and must be recovered before transfers resume.

Formalizing the Rule

- ▶ If we assume a target nominal net worth $\bar{N}^C$, the rule can be written as:

$$T_t^C = \begin{cases} \max(\Psi_t^C, 0) & \text{if } N_{t-1}^C \geq \bar{N}^C \\ 0 & \text{otherwise} \end{cases}$$

- ▶ **Result:** Under this asymmetric rule, open market operations remain **relevant** (affecting prices) if losses are large enough.

Summary: Helicopter Money & UMP

Comparing Policy Tools

- ▶ Our analysis of Helicopter Money and UMP converges on a single insight: effectiveness depends on generating **wealth effects**.

Mechanisms of Wealth Transfer

- ▶ **Helicopter Money:** Effectiveness relies on a *direct* fiscal transfer to households that is not negated by future tax increases.
- ▶ **UMP / QE:** Effectiveness relies on an *implicit* transfer. Asset purchases work only if the associated risk/wealth transfer behaves like a fiscal injection (i.e., central bank losses are not taxed back).

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Beyond Static Policy Analysis

Limitations of Previous Analysis

- ▶ Our analysis so far assumed a **perfect alignment** between the duration of the exogenous negative shock and ZLB policy.
- ▶ This simplification obscured the importance of interest rate policy *during* and *after* the crisis. The duration of zero interest rate policy beyond the occurrence of the shock can be an instrument of policy.
- ▶ Explicitly offering guidance on the future path of the policy rate is known as *Forward Guidance*.

Forward Guidance at the Zero Lower Bound

Strategic Relevance in a Liquidity Trap

- ▶ While Forward Guidance is a standard tool of monetary policy, it becomes critical when the short-term nominal interest rate binds at the Zero Lower Bound (ZLB).
- ▶ In a dynamic setting, the duration of the zero-interest rate policy is not strictly determined by the duration of the exogenous negative shock.
- ▶ Instead, the duration of the ZLB period becomes a **strategic choice variable** for the policymaker.
- ▶ **Mechanism:** Committing to keep rates at zero *longer* than the shock persists generates the inflationary expectations necessary to lower real interest rates and stimulate demand.

Optimal Policy Constraints and the ZLB

Revisiting Optimal Policy

- ▶ In the standard optimal policy analysis (Chapter 7), the Aggregate Demand equation was **not** a binding constraint.
- ▶ The framework implicitly assumed that the nominal interest rate $\hat{i}_t$ could freely adjust to track the natural rate r_t^e and the output gap x_t :

$$\hat{i}_t = r_t^e - (\sigma^{-1} - \theta^{-1})(1 - \gamma_1)x_t.$$

- ▶ *Validity Condition:* This solution remains valid only if shocks to r_t^e are sufficiently small, such that $\hat{i}_t \geq 0$ is naturally satisfied.
- ▶ In the presence of large negative shocks to the real rate r_t^e , the unconstrained solution violates the zero lower bound.
- ▶ Consequently, the optimization problem must be reformulated to explicitly incorporate:
 - 1 The Aggregate Demand (AD) equation as a binding restriction.
 - 2 The Zero Lower Bound (ZLB) inequality constraint, $\hat{i}_t \geq 0$.

The Constrained Lagrangian Formulation

Augmented Lagrangian from Chapter 7

$$\begin{aligned}
 \mathcal{L}_{t_0} = E_{t_0} \sum_{t=t_0}^{\infty} \beta^{t-t_0} & \left[\underbrace{\frac{1}{2}(\hat{Y}_t - \hat{Y}_t^e)^2 + \frac{1}{2} \frac{\theta}{\kappa} (\pi_t - \pi)^2}_{\text{Loss Function}} \right. \\
 & + \underbrace{\varphi_t [(\hat{Y}_t - \hat{Y}_t^e) - (\hat{Y}_{t+1} - \hat{Y}_{t+1}^e) + \sigma(\hat{i}_t - (\pi_{t+1} - \pi) - r_t^e)]}_{\text{AD Constraint}} \\
 & + \underbrace{\lambda_t [(\pi_t - \pi) - \kappa(\hat{Y}_t - \hat{Y}_t^e) - v_t - \beta(\pi_{t+1} - \pi)]}_{\text{AS Constraint}} \Big] \\
 & - \underbrace{\varphi_{t_0-1} \beta^{-1} (\hat{Y}_{t_0} - \hat{Y}_{t_0}^e) - \varphi_{t_0-1} \beta^{-1} \sigma(\pi_{t_0} - \pi) - \lambda_{t_0} (\pi_{t_0} - \pi)}_{\text{Initial Conditions (Timeless perspective constraints)}}
 \end{aligned}$$

Note: All variables and notations follow the definitions established in Chapter 7.

Optimality Conditions

First-Order Conditions (FOCs)

- ▶ Minimizing the Lagrangian with respect to inflation π_t and output gap $\hat{Y}_t$ yields:

$$\theta(\pi_t - \pi) - \beta^{-1}\sigma\kappa\varphi_{t-1} + \kappa\lambda_t - \kappa\lambda_{t-1} = 0 \quad (9.25)$$

$$(\hat{Y}_t - \hat{Y}_t^e) + \varphi_t - \beta^{-1}\varphi_{t-1} - \kappa\lambda_t = 0 \quad (9.26)$$

The Kuhn-Tucker Condition

- ▶ The non-negativity constraint on the nominal interest rate ($i_t \geq 0$) implies the complementarity condition:

$$i_t \cdot \varphi_t = 0 \quad \text{and} \quad \varphi_t \geq 0$$

- ▶ $\varphi_t > 0$ only when the ZLB is binding ($i_t = 0$). If not binding ($\varphi_t = 0$), we recover the standard solution of Chapter 7.

Synthesizing a Target Rule

Defining the "Adjusted Price"

- ▶ Combining the FOCs to express the optimal policy in terms of the target variable $\tilde{p}_t$ be the "adjusted price level" (price level + weighted output gap):

$$\tilde{p}_t = p_t + \theta^{-1}(\hat{Y}_t - \hat{Y}_t^e)$$

The Targeting Strategy

- ▶ The central bank aims to set $\tilde{p}_t$ equal to a target $\bar{p}_t^*$.
- ▶ **If feasible** ($i_t > 0$): The target is met perfectly: $\tilde{p}_t = \bar{p}_t^*$.
- ▶ **If infeasible** ($i_t = 0$): The actual $\tilde{p}_t$ falls below $\bar{p}_t^*$, and the gap accumulates in the target variable.

Optimal Policy

Law of Motion for the Target $\bar{p}_t^*$

- ▶ The target $\bar{p}_t^*$ is updated following the law of motion:

$$\bar{p}_{t+1}^* = \pi + \bar{p}_t^* + \beta^{-1}(1 + \kappa\sigma)(\bar{p}_t^* - \tilde{p}_t) - \beta^{-1}(\bar{p}_{t-1}^* - \tilde{p}_{t-1}).$$

- ▶ This relation is obtained by eliminating the Lagrange multipliers and defining the target as $\bar{p}_t^* \equiv \varphi_t + \tilde{p}_t$.
- ▶ The target is adjusted upward during periods in which it is not achieved ($\bar{p}_t^* > \tilde{p}_t$) and the zero-lower bound is tight. This increase in the target during the liquidity trap works to push up inflation expectations.
- ▶ Conversely, when the zero-lower bound is no longer binding, the price target decreases at that moment, as captured by the last term on the RHS.
- ▶ Under normal conditions of a positive nominal interest rate: $\tilde{p}_t = \bar{p}_t^*$ and the target simply grows at the rate π (as discussed in Chapter 7).

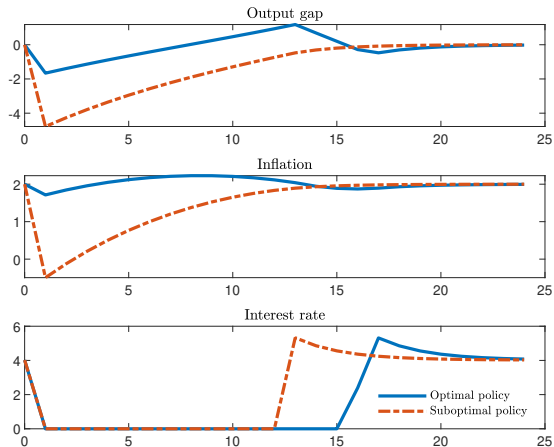
Comparison: The Suboptimal Policy

- ▶ Consider a central bank that **never adjusts** its target price level trajectory, even after missing it.
- ▶ The target follows the standard path $\tilde{p}_{t+1}^* = \pi + \tilde{p}_t^*$ (as in Chapter 7).

Implementation and Limitation

- ▶ **Implementation:** If feasible ($i_t > 0$), the target is met ($\tilde{p}_t = \tilde{p}_t^*$). If infeasible ($i_t = 0$), the central bank simply accepts the deviation.
- ▶ **The Flaw:** While optimal *outside* the ZLB, this strategy is suboptimal when binding because it fails to "catch up" to the lost price level trajectory.

Simulation: Optimal vs. Suboptimal Policy



Responses of output gap, inflation, and interest rate to a negative shock to the natural rate of interest (duration: 12 quarters). **Solid Line:** Optimal Policy. **Dashed Line:** Suboptimal Policy (Standard Targeting). Variables are in %. Rates are annualized. Parameters are calibrated as in Chapter 7.

Key Result: "Lower for Longer"

The Optimal Response

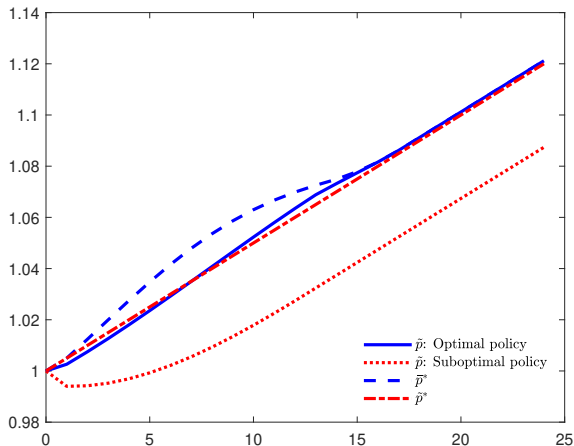
- ▶ The economy is hit by a severe negative shock: the natural rate falls to -4% (annualized) for 12 quarters.
- ▶ A suboptimal (standard) policy would keep rates at zero for exactly 12 quarters, rising immediately as the shock subsides.
- ▶ The optimal policy keeps the interest rate at zero for 15 quarters.
- ▶ This captures the essence of *Forward Guidance*: By committing to keep rates low even after the natural rate recovers, the central bank generates the inflation expectations needed to stimulate the economy *today*.
- ▶ **Implication:** Another way to communicate Forward Guidance is to explicitly promise **higher inflation than the target** (overshooting) after the negative shock ends.

Forward Guidance in Practice

Central Bank's have progressively adopted more specific forms of guidance to implement these theoretical insights:

- 1 **Qualitative (Fed 2003):** "Policy accommodation can be maintained for a *considerable period*."
- 2 **Extended (Fed 2008):** Rates will stay low for an "*extended period of time*."
- 3 **Time-Contingent (Fed 2011):** Specific dates provided (e.g., rates will stay low "at least through mid-2013").
- 4 **State-Contingent (Fed 2012):** Lift-off linked to specific economic outcomes (unemployment > 6.5% and stable inflation), rather than calendar dates.

Price Level Dynamics



Evolution of the adjusted price level $\tilde{p}_t$ under Optimal Policy vs. Suboptimal Policy. The graph compares the realized path against the optimal target $\tilde{p}_t^*$ and the suboptimal target $\tilde{p}_t^*$.

Price Level under Optimal vs. Suboptimal Policies

Suboptimal Policy: “Bygones are bygones”

- ▶ The adjusted price level $\tilde{p}_t$ (dotted red line) falls sharply during the initial liquidity trap. Once the constraint lifts, the price level resumes growing at the standard 2% rate, but the permanent shortfall is accepted — the central bank ignores past misses.

Optimal Policy: History Dependence

- ▶ The optimal path $\tilde{p}_t$ (solid blue line) attempts to catch up with the target $\bar{p}_t^*$ (dashed blue line). To achieve this catch-up, inflation must eventually exceed 2%.

Policy Framework Reviews

- ▶ The implications of this New-Keynesian model directly support the strategic reviews conducted by major central banks.

Federal Reserve (2020 Review)

- ▶ Adopted **Average Inflation-Targeting**.
- ▶ Explicitly acknowledges that periods of inflation persistently below 2% should be followed by periods of inflation *above* the target.

European Central Bank (2021 Review)

- ▶ Replaced "below but close to 2%" with a **symmetric 2% target**.
- ▶ Explicitly recognizes the asymmetry imposed by the effective lower bound. This strategy allows for "a temporary period during which inflation moderately exceeds the target", consistent with the optimal policy prescription.

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Beyond Interest Rate Policy: Asset Purchases

The Challenge of "Tapering"

- ▶ While Forward Guidance clarifies the future path of interest rates, the path and timing of asset purchases (QE) represent a distinct policy dimension.
- ▶ **Historical Context:** The "Taper Tantrum" of June 2013 illustrates the market's sensitivity to these flows.
- ▶ When the Federal Reserve signaled a potential reduction in the pace of asset purchases ("tapering"), markets reacted with panic, triggering a sharp spike in U.S. Treasury yields.

The Mechanics of QE: The Reserves Channel

Alternative Transmission Mechanisms

- ▶ To analyze QE theoretically, we focus on the **Reserves Channel** (the "Monetarist View"), where an increase in central bank reserves is supposed to trigger liquidity creation via financial intermediaries, thereby stimulating aggregate demand.
- ▶ We use the New-Keynesian framework with financial frictions introduced in Chapter 8.
- ▶ In this setup, reserves are not neutral. They provide distinct non-pecuniary benefits to intermediaries by relaxing collateral or regulatory constraints.

Optimal Liquidity Policy and Satiation

Applying the ZLB to the Banking Model

- ▶ We apply the zero-lower bound constraint to the optimal policy problem from Chapter 8. Strikingly, the optimal liquidity policy remains **unchanged** relative to the unconstrained case.
- ▶ The central bank should supply reserves up to the point of **satiating** the liquidity needs of the economy.
- ▶ Once the economy reaches the satiation level, further increases in central bank reserves are **irrelevant**.
- ▶ **Conclusion:** Reserve policy is impactful only when satiation is *not* yet reached.

Liquidity with Distortionary Taxation

The Assumption of Costly Liquidity

- ▶ The role and dynamic of liquidity become more significant when it is costly to vary it, such as in a context in which lump-sum taxes are not available.
- ▶ We reconsider the model of Chapter 8 under the assumption that the only available tax is the distortionary tax on firm's revenues (τ).
- ▶ Under this assumption, the overall supply of government liquidity, B_{t-1}^g , is directly related to the distortionary tax through the intertemporal resource constraint of the economy.

The Intertemporal Resource Constraint

- ▶ The resource constraint linking liquidity and fiscal variables, replacing (8.19), is given by:

$$\frac{(1 + i_{t-1}^X)B_{t-1}^g}{P_t} = E_t \left\{ \sum_{T=t}^{\infty} R_{t,T} \left[\tau_T Y_T - Tr_T + \frac{i_T - i_T^X}{1 + i_T} \frac{B_T^g}{P_T} \right] \right\}. \quad (9.27)$$

Variable Definitions

- ▶ $R_{t,T}$ is the stochastic discount factor for real income.
- ▶ i^X is the policy rate.
- ▶ i is the market nominal interest rate.
- ▶ Tr is an exogenous non-negative transfer.
- ▶ τ is the distortionary tax on firms' revenues.

Welfare Implications of Costly Liquidity

The Trade-off

- ▶ Supplying liquidity generates taxation-related distortions that can negatively impact overall welfare.
- ▶ As previously discussed in Chapter 8, the optimal supply of liquidity is therefore deemed to be **below the satiation level**.
- ▶ Increasing liquidity beyond this point would necessitate significant distortions in taxation that outweigh the benefits of additional liquidity services.
- ▶ We analyze the optimal government stabilization problem using a linear-quadratic approach.
- ▶ The approximations are centered around an optimal steady state where inflation is at the target level, but liquidity remains below the satiation threshold.

The Quadratic Loss Function

- The quadratic welfare-based loss function takes the following form:

$$L_{t_0}^{CB} = E_{t_0} \sum_{t=t_0}^{\infty} \beta^{t-t_0} \left\{ \frac{1}{2} \Lambda_x x_t^2 + \frac{1}{2} \Lambda_{\pi} (\pi_t - \pi)^2 + \underbrace{\frac{1}{2} \Lambda_d \hat{d}_t^2}_{\text{Real Liquidity}} \right\} \quad (9.28)$$

- The parameters Λ_x , Λ_{π} , and Λ_d are positive weights derived from structural parameters (G. Benigno and P. Benigno 2022).
- The policymaker aims to minimize the deviation of the output gap (x), inflation (π), and real liquidity (deposits, $\hat{d}$) from their steady-state values, given the supply, demand and resource constraint.
- The analysis considers two stochastic disturbances: the preference shock ξ and the exogenous transfer Tr .

Modified Supply Constraint

The Aggregate Supply (AS) Equation

- ▶ The optimal policy problem is subject to a modified Phillips Curve (relative to Chapter 8):

$$(\pi_t - \pi) = \kappa \left[x_t + \underbrace{\psi_\tau(\tilde{\tau}_t - \tilde{\tau}_t^*)}_{\text{Tax Distortion Effect}} \right] + \beta E_t(\pi_{t+1} - \pi)$$

Interpretation

- ▶ This equation accounts for the time-varying supply-side effects of the distortionary tax on firms' revenue ($\psi_\tau > 0$).
- ▶ $\tilde{\tau}_t = \tau_t - \tau$ represents the deviation of the tax rate from its steady state.
- ▶ $\tilde{\tau}_t^*$ represents the "efficient" tax deviation—a combination of shocks such that if $\tilde{\tau}_t = \tilde{\tau}_t^*$, the tax wedge is neutralized, allowing output and inflation to be stabilized at their targets.

Modified Demand and Fiscal Constraints

Aggregate Demand (AD) with Banking Frictions

$$x_t = v_\rho E_t x_{t+1} - \sigma v_\rho (\hat{t}_t^X - E_t(\pi_{t+1} - \pi) - r_t^e) + d_y^{-1}(1 - v_\rho)\hat{d}_t \quad (9.29)$$

- ▶ r_t^e is the **constrained efficient real rate** (function of shocks).
- ▶ Parameters v_ρ, σ, d_y are positive.

The Linearized Resource Constraint (9.27)

$$\begin{aligned} \hat{d}_{t-1} - (\pi_t - \pi) - \sigma^{-1}x_t + (\hat{t}_{t-1}^X - r_{t-1}^e) = \\ -f_t + E_t \sum_{T=t}^{\infty} \beta^{T-t} \left[b_x x_T + b_\tau (\tilde{\tau}_T - \tilde{\tau}_T^*) + b_d \hat{d}_T \right] \end{aligned} \quad (9.30)$$

Feasibility and Fiscal Stress

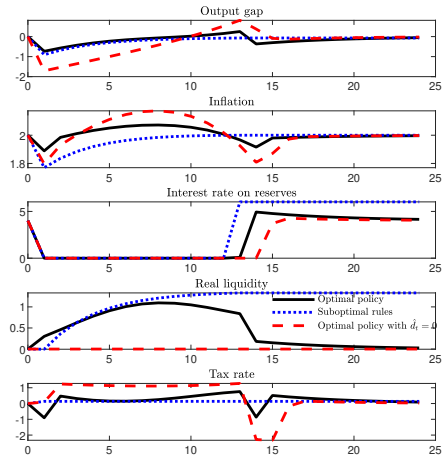
Eggertsson and Woodford (2004): Fiscal Stress (f_t)

- ▶ The variable f_t measures the "**Fiscal Stress**": the extent to which full stabilization of targets (x, π, d) is incompatible with the government's intertemporal budget constraint.
- ▶ If $f_t > 0$, the government lacks the fiscal space for a constrained optimal outcome.

Feasibility Conditions

- ▶ **Ideal Case** ($f_t = 0$): It is feasible to reach all three targets provided the natural rate does not violate the ZLB. In this case, the policy rate tracks the natural rate perfectly: $\hat{i}_t^X = r_t^e$.
- ▶ **The ZLB Constraint**: However, if r^e falls substantially, the ZLB binds ($i^X \geq 0$). Even without fiscal stress, this forces a trade-off between stabilizing the relevant variables.

Comparison of Policy Simulations



Comparison between the optimal policy, a sub-optimal policy, and the optimal policy with no adjustments to liquidity.

Suboptimal Policies

Benchmark 1: Constrained Inflation Targeting

- ▶ **Monetary Rule:** Strictly targets inflation ($\pi_t = \pi$). If the target requires $i_t < 0$, the rate hits the zero floor.
- ▶ **Fiscal Role:** *Passive*. Maintains a constant expected tax gap consistent with solvency; does not react to the crisis.

Benchmark 2: Constant Liquidity Policy

- ▶ **Objective:** Isolates the optimal response **without** active liquidity management.
- ▶ **Fiscal Rule:** Adjusts taxes purely to stabilize real liquidity at the steady state ($\hat{d}_t = 0$).
- ▶ **Monetary Rule:** Minimizes the loss function subject to this liquidity-neutral fiscal constraint.

Analysis: Optimal vs. Suboptimal Outcomes

The Cost of Suboptimal Policies

- ▶ **Deep Contraction:** Significant welfare losses characterized by a sharp drop in output and inflation persistently below target.
- ▶ **Premature Liftoff:** The policy rate rises from the ZLB **exactly** when the shock vanishes. This lack of commitment fails to stimulate expectations.

The Gains from Optimization

- ▶ **Stabilization:** Optimal policy leverages multiple tools to keep inflation close to target and output variations moderate.
- ▶ **Mechanism:** It creates a "history-dependent" path that suboptimal rules ignore.

Feature 1: Optimal Duration and Liftoff

Forward Guidance: “Lower for longer”

- ▶ As discussed in the Forward Guidance section, the optimal policy keeps the interest rate at zero for longer than the duration of the shock itself.
- ▶ Although the optimal policy remains low longer than a *standard* rule, it lifts off *earlier* than under a *constant liquidity policy*.
- ▶ This highlights the benefits and complementarities of actively managing reserves and liquidity.

Feature 2: Dynamics of Asset Purchases (QE)

Reversibility vs. Permanence

- ▶ **Expansion:** Liquidity is injected aggressively at the start of the trap, peaking just before the shock ends.
- ▶ **Normalization:** A sharp withdrawal occurs precisely at lift-off, followed by a gradual return to the steady state as the balance sheet is fully unwound (reversibility).
- ▶ **Sub-optimal policy:** A *permanent* accumulation of public debt increases long-run tax distortions.

Feature 3: Optimal Fiscal Coordination

- ▶ Optimal policy requires the distortionary tax rate (τ) to adjust at two critical moments:

1. At the Onset (Producing Fiscal Stimulus)

- ▶ Taxes are **lowered** to stimulate the rise in government liquidity.

2. At the Exit (Muting Overshooting of Inflation)

- ▶ Taxes are **lowered again** at the moment of liftoff, since lower taxes reduce marginal costs (via the *AS* equation).
- ▶ Active liquidity policy reduces the need of inflation overshooting at the exit of the liquidity trap.

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Rediscovering the Lender of Last Resort

The responses to the 2007-2008 financial crisis and the COVID-19 pandemic extended well beyond standard macroeconomic stabilization. They revived the central bank's foundational role.

The Original Mandate (1913)

- ▶ The Federal Reserve was established specifically to address banking panics.
- ▶ President Woodrow Wilson signed the Federal Reserve Act to grant authority not just for monetary policy, but to contain systemic risk and maintain financial stability.

The Role of the Central Bank

- ▶ As the monopoly issuer of fiduciary currency and risk-free liabilities, the central bank is uniquely positioned to prevent the illiquidity of intermediaries from devolving into insolvency.

From Bagehot to "Buyer of Last Resort"

Bagehot's Dictum (1873)

- ▶ *Unlimited Lending*: Provide liquidity to any entity in the economy.
- ▶ *Penalty Rate*: Loans should carry a premium over the risk-free rate to discourage moral hazard.
- ▶ *Collateral Valuation*: Crucially, collateral must be evaluated at "normal" prices. Valuing assets under stress would reflect panic-induced discounts, failing to distinguish between illiquidity and insolvency.

Modern Evolution

- ▶ While the discount window serves as the traditional tool, the complexity of modern crises necessitated an evolution.
- ▶ Central banks have effectively become *buyers of last resort* (Bernanke 2022), stepping in when lending alone is insufficient to stabilize markets.

Mitigating Systemic Risk and Fire Sales

The Vicious Cycle of Financial Stress

- ▶ During severe stress, intermediaries face deteriorating asset values and stricter margin requirements.
- ▶ This raises funding costs, often triggering forced liquidation of assets.
- ▶ *Systemic Consequence:* Simultaneous liquidations cause "fire sales" depressing prices further across the entire market and exacerbating the funding crunch.

Policy Intervention

- ▶ By purchasing assets directly or accepting stressed collateral at favorable terms, the central bank breaks this spiral.
- ▶ As detailed in Chapter 10, these policies mitigate the adverse feedback loops between financial friction and aggregate demand.

The 2008 Financial Crisis

Shadow Banking

- ▶ A major complexity of 2008 was the shadow banking system, which was ineligible for traditional discount window lending.

The Response: Section 13(3)

- ▶ Invoking "extraordinary circumstances" under the Federal Reserve Act, the Fed extended lending outside the traditional banking system.
- ▶ *Commercial Paper Funding Facility (CPFF)*: Provided direct liquidity to corporations.
- ▶ *Term Asset-Backed Securities Loan Facility (TALF)*: Supported investors purchasing credit securities.
- ▶ *Money Market Mutual Fund Liquidity Facility (MMLF)*: Lent to banks against collateral purchased from prime money market funds.

The COVID-19 Pandemic

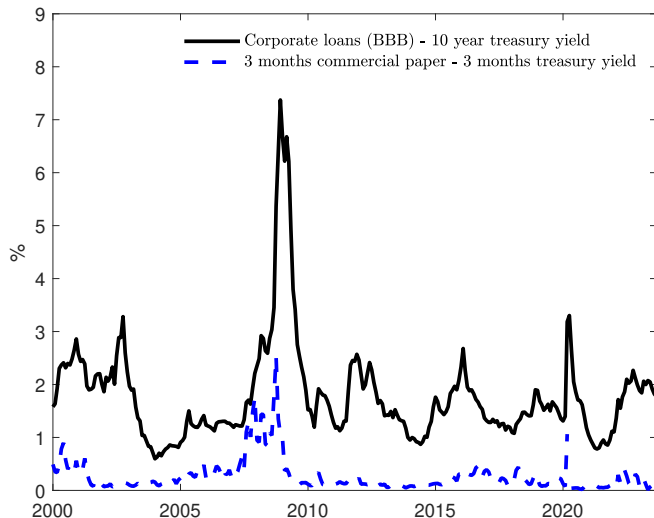
From MainStreet to Wall Street

- ▶ Unlike 2008, the pandemic shock originated in the real economy ("Main Street") before extending to the financial sector.
- ▶ Businesses faced an immediate liquidity crisis regarding fixed costs and payrolls due to lockdowns.

Expanding the Safety Net

- ▶ The Fed reactivated the 2008 facilities and launched new programs targeting small and mid-sized entities.
- ▶ *Main Street Lending Program*: Targeted support for SMEs (April 2020).
- ▶ *Paycheck Protection Program (PPP)*: Crucial for maintaining workforce attachment.
- ▶ These interventions were instrumental in preventing a wave of foreclosures and bankruptcies.

Evidence: The Impact on Credit Spreads



- ▶ The figure illustrates the dramatic surge in credit spreads during both the 2008 crisis and the COVID-19 pandemic.
- ▶ It also demonstrates the central bank's ability to compress spreads and restore market functioning.

Looking Ahead: The Safe Asset Shortage

The Liquidity Crunch Scenario

- ▶ Chapter 11 will explore a scenario where the economy faces a critical shortage of "safe assets"—debts that serve as collateral or reliable stores of value.
- ▶ The 2008 crisis revealed that private "money-like" securities (e.g., within the shadow banking system) lose their liquidity services exactly when needed most.
- ▶ The central bank can counter the shortage of "safe assets" during a liquidity crisis by expanding reserves.

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Main Takeaways: Concepts and Tools

The liquidity trap

- ▶ Occurs when negative demand shocks interact with price rigidity and the ZLB.
- ▶ The result is a persistent shortfall in aggregate demand that cannot be cured by relying only on standard interest-rate cuts.

Helicopter money

- ▶ Fiscal transfers financed by a permanent expansion of monetary liabilities.
- ▶ **Mechanism:** Raises long-run price expectations and stimulates consumption via the wealth effect, provided the liability increase is not offset by future taxes.

Unconventional monetary policy (QE)

- ▶ Pure asset swaps are neutral in a frictionless model.
- ▶ To be effective, QE must induce a wealth transfer to the private sector—implying that the central bank or the Treasury absorbs risk and losses.

Main Takeaways: Implementation Strategy

Forward guidance

- ▶ Promises to keep rates at zero *beyond* the duration of the shock.
- ▶ **History dependence:** allowing inflation to temporarily overshoot (“catch up”) lowers real interest rates and mitigates the recession.
- ▶ This approach allows the price level to reconnect with its trend in the long run.

Managing reserves

- ▶ When reserves relax banking constraints, they should be increased aggressively during the trap and withdrawn gradually at liftoff.
- ▶ Fiscal policy (taxes) must adjust in a way that complements the liquidity policy.

Financial stability

- ▶ The central bank acts as the buyer of last resort during systemic crises.
- ▶ Targeted credit-easing operations prevent fire sales and stabilize spreads, complementing aggregate-demand management.

9 Appendix

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Dynamics of central bank Net Worth

- For the constant price level P^* to constitute an equilibrium, central bank net worth must remain time-invariant. We analyze the evolution of net worth:

Law of Motion

$$N_t^C = N_{t-1}^C + \Psi_t^C - T_t^C$$

- Using the definition of net worth (9.23), profits represent the sum of the risk-free return on net assets and the risk premium earned on the bond portfolio:

$$\Psi_t^C = \underbrace{i_{t-1}(N_{t-1}^C + M_{t-1})}_{\text{Risk-free return}} + \underbrace{(i_t^L - i_{t-1})Q_{t-1}^L \tilde{B}_{L,t-1}^C}_{\text{Excess return on Long-Term Bonds}} \quad (9.31)$$

- Substituting (9.31) into the law of motion:

$$N_t^C = (1 + i_{t-1})N_{t-1}^C + i_{t-1}M_{t-1} + (i_t^L - i_{t-1})Q_{t-1}^L \tilde{B}_{L,t-1}^C - T_t^C \quad (9.32)$$

Proof of Net Worth Stability

- ▶ Substituting the transfer rule (9.20) into (9.32).
- ▶ The variable components (profits/losses) cancel out, yielding:

$$N_t^C = (1 + i_{t-1})N_{t-1}^C - (1 - \beta)P_t\tau^C$$

or

$$N_t^C = N_{t-1}^C + P_t \left[\underbrace{i_{t-1} \frac{N_{t-1}^C}{P_t} - (1 - \beta)\tau^C}_{= 0} \right]$$

Conclusion

- ▶ From the equilibrium condition (9.22) and since $i_{t-1} = \frac{1}{\beta} - 1$, the term in brackets is identically zero, **net worth is constant** at the initial level: $N_t^C = N_{t-1}^C = N_{t_0-1}^C$.

[Back to Equilibrium Price Level Stability](#)

9 Appendix

10 Bibliography

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